

**RECORDED THE
GROWTH OF
MORE THAN
29,000
PEA PLANTS**



**WAS A
MEMBER OF
34
SCIENTIFIC
SOCIETIES**

second generation contained yellow-seeded and green-seeded plants in the exact ratio of 3:1. Mendel saw that "either/or" traits were passed on in exact proportions. He called this the "law of segregation." He also realized that these characteristics were determined by discrete "particles," now called genes, that occurred in pairs and were formed at fertilization because one gene was inherited from each parent plant.

Dominant and recessive genes

As Mendel discovered, there are different versions of a gene for any particular trait. Some are stronger ("dominant") and some weaker ("recessive"). Dominant versions override recessive ones, so when different versions are inherited, only the dominant trait appears in the next generation. With Mendel's peas, the gene for yellow seeds was dominant. As the offspring of his first cross inherited one "yellow" gene and one "green" gene, all produced yellow seeds. But when two of these plants were crossed, one-quarter of their offspring would inherit

a "green" (recessive) gene from each parent and have green seeds, while the rest were yellow. This hereditary mechanism explains why some traits can skip generations: for example, a red-haired parent might have a dark-haired child but a red-haired grandchild.

It was fortunate that Mendel studied peas, as their genetic mechanism is very simple. The process in many other organisms is more complex, although still based on the same rules. Most characteristics are governed by a combination of genes, resulting in a more variable outcome (height in humans, for example). Moreover, some genes for different traits are linked together and are therefore inherited together.

Mendel published his findings, but they were ignored for more than 30 years. Around 1900, they were rediscovered, and he received the credit he deserved.



In his pea-breeding experiments, Mendel found that some genes could be carried without being expressed as physical traits. In this pea "family tree," the first generation all carry the gene for red seeds, and the red-seeded trait becomes visible when the genes recombine in the second generation.

HUGO DE VRIES

The Dutch botanist de Vries independently reached the same conclusions as Mendel about heredity, and brought genetics to the attention of the scientific world.



The young de Vries (1848–1935) was passionate about plants and was also an admirer of Darwin's theory of evolution. Intrigued to learn more about how evolution worked, he carried out plant-breeding experiments on evening primrose and used the word "mutation" for the spontaneous variations that sometimes occurred. Further experiments led him to the same results as Mendel, which he published in 1900. De Vries also coined the term "pangene" for the unit of heredity, which later became "gene."